Trust, Verified: Understanding and Engineering Trustworthy Private Systems

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My research sits at the intersection of cryptography and policy, where cryptographic guarantees meet the messy world of social, legal, and regulatory challenges. I use three complementary methods: **applied cryptography research** to design and implement new constructions that extend the frontier of cryptography applications; **empirical work** conducting security evaluations and user studies of deployed systems to understand the status quo; and **policy engagement** with government agencies, standards bodies, and civil society organizations to bridge the gap between technical findings and policy impacts. These methods support my research on **privacy-enhancing and verifiable technologies, especially for digital identity, age verification, and messaging**.

Digital tools like messaging applications and digital IDs can improve users' lives through convenience, privacy, and security—but an insecure or untrustworthy application can instead expose sensitive information or deny access to core services. How can users and their devices know whether an application is helping them or hurting them? This question is especially urgent for digital identity and attribute verification. Recent U.S. age-verification laws increasingly require online checks using biometrics or IDs, mobile IDs are planned in more than half of U.S. states, and the EU is planning to release its Digital Identity Wallet by the end of 2026. These systems promise stronger checks against fraud and can even improve users' privacy compared to full physical ID disclosure, but an untrustworthy digital ID application would be a disaster for users' privacy, security, and autonomy. My work aims to understand these systems, find their failures, and build new trustworthy ones.

My empirical work characterizes the systems being deployed today. In an n=1,635 web experiment—the first to measure behavior when adults encounter age verification incidentally—we found that completion fell from 99% for self-attestation to 18–27% for government-ID methods indicating deep user aversion to the system. Participants cited distrust of providers, identity-theft risks, and loss of anonymity as primary points of concern [LZXCS; USENIX'26]. I also analyzed the two-way relationship between technical architecture and requirements in age verification laws [LS; CS&Law'26], and I recently evaluated deployed biometric age-assurance models for bias and accuracy in adversarial settings [CXWMSF and XHCSM; under review at USENIX'27].

Beyond studying deployed systems, I build new cryptographic mechanisms that provide, communicate, and verify strong privacy guarantees where they are needed most. End-to-end encrypted messaging is one such setting: encryption protects messages from providers, but makes it difficult to conduct content moderation; moreover, tools designed to detect child sexual abuse material can be silently repurposed to surveil lawful speech. I designed cryptographic checks that let users' devices detect such repurposing and failures [SKM; IEEE S&P'23], and built a system that lets journalists infer trends from encrypted messages using differential privacy and secure multi-party computation [XJGS; manuscript'25]. I have also analyzed the governance of hash sharing databases for terrorist content [SS, DfP Europe'25], court-compelled decryption [SV, USENIX'21], and broader issues in content moderation under encryption [SM, PETS'23, SM, PETS'24].

Beyond publication, I actively work to inform real-world policy by sharing findings at policy venues. I have spoken at the Federal Trade Commission, State of the Net, standardization meetings, industry events, and discussions with state lawmakers and attorneys general.

My priority going forward is to bring technical user-verifiable checks to ID applications so that users can have the benefits of *trustworthy* digital ID systems. As my focus expands from age assurance to full ID verification, I am building zero-knowledge processing of physical-ID barcodes into digital credentials [MWZS; manuscript'26]; I am also analyzing deployed state digital-ID applications to identify discrepancies among their privacy notices, specifications, and code [LCS; planned for submission to CS&Law'27].

References

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